

MX17 DREAM DAQ Returned to CEA Saclay

Oct 5, 2026 · Dylan Neff & Stephan Aune

The MX17 DREAM DAQ was returned to CEA on 5 October 2026 and tested the same day: every FEU powered up, communicated with the TCM and produced normal pedestals.

Background

The DAQ was assembled in 2026 for the MX17 Micromegas tracker at CERN n_TOF, where it read out the X17 search in July and August 2026. It was returned on 5 October 2026 by Jean-Yves Rousse, together with everything else borrowed for the campaign, including all DREAM cables and other small items.



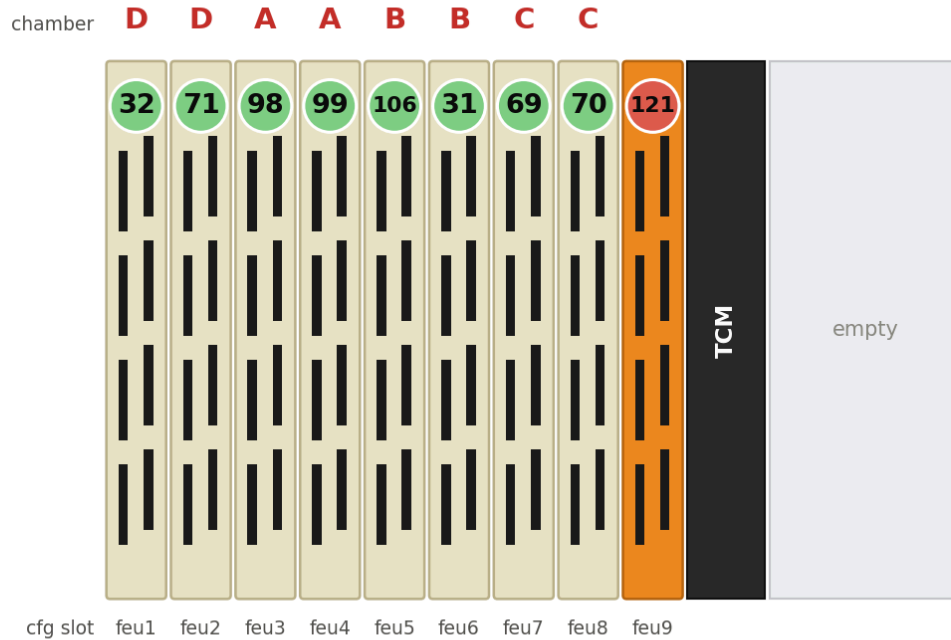
Figure 1. The MX17 DREAM DAQ crate at CEA Saclay on 5 October 2026: the nine FEUs and the TCM in the top crate, the CAEN unit below.

Test

1. **Power-on.** Everything was plugged in and switched on; all units turned on.
2. **One FEU at a time.** The low-voltage supply used at n_TOF was rented from CERN and went back there, so a smaller 10 A LV unit was used. It can power the TCM and only one FEU at once, so each of the 9 FEUs was tested in turn with the TCM.
3. **Pedestals, no load.** With no detector cables plugged in, a 1000-event pedestal run was taken on each FEU (13:27–13:40). This checks that each FEU communicates as expected once powered, and that every DREAM ASIC gives a reasonable pedestal.



Figure 2. Close-up of the FEUs. Each card carries its FEU ID on a round sticker near the top; the tape above marks the chamber each pair read at n_TOF.

MX17 DREAM crate, front view — FEU IDs left to right

ID stickers as on the cards (green); the orange card's red sticker is illegible in the photo, its ID is read from its data stream.

Figure 3. The same crate drawn with the IDs legible, left to right, with the configuration slot each was tested as. The orange card's red sticker cannot be read in the photo; its ID, 121, is the one in its data stream.

Results

All 9 FEUs and all 4608 channels (72 DREAM ASICs) gave normal pedestals.

- **Communication:** each FEU was read out on its own, and the FEU ID in its data stream matched the ID the configuration assigns to it.
- **Data integrity:** 9,546 events in total, none missing and no lost samples.
- **Pedestals:** baselines between 194 and 453 ADC on every channel.
- **Noise:** median 3.0 ADC raw and 2.4 ADC after common-noise subtraction, equal across FEUs to within 3 %. These low values are expected with no detector attached.

MX17 DREAM DAQ back at CEA — pedestals on all 9 FEUs, 5 October 2026

4608 channels, 9,546 pedestal events, one FEU at a time · no channel more than 2× or less than ½× its FEU's median noise



Figure 4. Every channel of the 9 FEUs side by side. Top: pedestal baseline. Bottom: noise before (orange) and after (blue) common-noise subtraction. The repeating pattern is the 64-channel structure of each DREAM ASIC, identical on every card.

Conclusion

Everything looks normal. The DAQ was returned to CEA Saclay on 5 October 2026 and verified to be in working condition on the same day.

Compiled with Claude (Anthropic)